

Loan Approval Prediction Report

InternSpark Internship – Machine Learning Task 2

1. Introduction

Loan approval is an important decision in the banking and financial sector. A bank normally reviews several details of an applicant before deciding whether a loan should be approved. In this project, machine learning was used to predict loan approval from borrower information. The work mainly focused on practical data preparation, class imbalance, model comparison and evaluation.

2. Problem Statement

The objective of this project is to build a supervised classification model that predicts whether a loan application will be approved or rejected. The target variable is `Loan_Status`, where Y represents an approved application and N represents a rejected application. The project also evaluates different decision thresholds so that the final prediction can be aligned with a business requirement.

3. Dataset Description

The dataset used is the Loan Approval Prediction dataset from Kaggle. It contains 614 records and 13 columns. The input information includes gender, marital status, dependents, education, employment status, applicant income, co-applicant income, loan amount, loan term, credit history and property area. `Loan_Status` is the target variable. There are 422 approved applications and 192 rejected applications.

4. Data Preprocessing

`Loan_ID` was removed because it is an identifier and does not provide useful predictive information. The target variable was converted from Y/N into 1/0. Numerical and categorical variables were handled separately. Numerical variables were imputed using the median and scaled with `StandardScaler`. Categorical variables were filled with the most frequent value and converted using one-hot encoding. A `ColumnTransformer` and `Pipeline` were used so that the same preprocessing was applied consistently during training and prediction.

5. Missing Value Handling

Missing values were present in Gender (13), Married (3), Dependents (15), Self_Employed (32), LoanAmount (22), `Loan_Amount_Term` (14) and Credit_History (50). Instead of deleting rows, numerical missing values were replaced with the median and categorical missing values with the most frequent category.

6. Class Imbalance Handling

The target classes were not evenly distributed, with 422 approved and 192 rejected applications. To reduce the tendency of a classifier to favor the larger class, `class_weight='balanced'` was used in both Logistic Regression and Random Forest. This approach kept all available observations.

7. Model Development

Logistic Regression was used as the baseline classification model because it is simple and comparatively easy to interpret. Random Forest was used as the tree-based model because it can capture nonlinear relationships and interactions between borrower features. The Random Forest model used 300 trees, balanced class weights, a maximum depth of 8 and a minimum split size of 5.

8. Model Evaluation

The data was split into training, validation and test sets using stratification. Precision, recall, F1-score and ROC-AUC were used for evaluation. At the default threshold of 0.50, Logistic Regression achieved precision 0.864, recall 0.797, F1-score 0.829 and ROC-AUC 0.834. Random Forest achieved precision 0.848, recall 0.875, F1-score 0.862 and ROC-AUC 0.825.

9. Model Comparison

There is no single model that wins on every metric. Logistic Regression was stronger on precision and ROC-AUC, while Random Forest was stronger on recall and F1-score. Since loan screening can benefit from identifying more potentially eligible applicants, Random Forest was selected for further threshold analysis.

10. Threshold Analysis

The default threshold of 0.50 was not treated as automatically optimal. Thresholds from 0.20 to 0.80 were evaluated on the validation set. A threshold of approximately 0.37 gave the best validation F1-score. Applying 0.37 to the unseen test set increased Random Forest recall from 0.875 to 0.938 and F1-score from 0.862 to 0.882, while precision changed from 0.848 to 0.833.

11. Feature Importance

Random Forest feature importance showed that Credit_History was the strongest feature, followed by LoanAmount, ApplicantIncome and CoapplicantIncome. These values describe the usefulness of features to the fitted model and should not be treated as proof of direct causation.

12. Business Interpretation

The model can be used as a screening or decision-support tool. Credit history was the strongest feature in this dataset, while income and requested loan amount also contributed strongly. The model can help prioritize applications for further review, but it should not replace the complete lending process, policy checks or human review.

13. Model Trade-offs

Logistic Regression is easier to explain and performed slightly better on precision and ROC-AUC. Random Forest is more flexible and achieved higher recall and F1-score. The trade-off is therefore between simpler interpretation and stronger balance on the selected classification metrics. For this internship dataset, Random Forest was preferred for the final threshold analysis.

14. Suggested Deployment Threshold

A threshold of approximately 0.37 is suggested for this project because it produced the best validation F1-score and improved test recall to about 93.8%. This setting is useful when missing a potentially eligible applicant is considered more costly than sending some additional applications for manual review. A real lending deployment should validate the threshold using larger and more recent data, business costs, risk policies and regulatory requirements.

15. Conclusion

This project developed a complete supervised learning workflow for loan approval prediction. Missing values, categorical variables, scaling and class imbalance were handled using a reproducible pipeline. Logistic Regression and Random Forest were compared using precision, recall, F1-score and ROC-AUC.

Random Forest gave the strongest F1-score at the default threshold and performed even better on F1 after threshold tuning. The project shows that model performance depends not only on the algorithm but also on preprocessing, class handling and the choice of decision threshold.

Model Performance

Model	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.864	0.797	0.829	0.834
Random Forest	0.848	0.875	0.862	0.825

Feature Importance – Random Forest

Feature	Importance
Credit_History	0.262
LoanAmount	0.167
ApplicantIncome	0.165
CoapplicantIncome	0.126
Loan_Amount_Term	0.050
Property_Area_Semiurban	0.041
Education_Not Graduate	0.032
Dependents_2	0.030
Married_Yes	0.029
Dependents_1	0.023